

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10 – ELECTROSTATICS (3.4) – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving Question: How do invisible electric charges build up, exert forces across empty space, and store energy, and how can we harness these abilities to protect communities and power technology in Kenya? In this Final Explanation you will use evidence from all nine lessons to explain the lightning strike at Ol Kalou Market. Write in your own words, in full sentences, in the space under each prompt. Use evidence from the experiments you did (balloons, combs, electroscopes, the foil capacitor) and name the physics ideas you use. Include labelled diagrams or equations where they help. Refer back to the Ol Kalou market stall, the wooden stall and the lightning conductor on the district offices in every section.

Part 1: How Charge Builds Up and Behaves on Different Materials

Before the strike, the traders at Ol Kalou saw their arm hairs stand on end and heard their transistor radio crackle. (a) Using what you know about atoms, explain how objects become electrically charged and what the two types of charge are. (b) State the law of charges and use it to explain why the traders' arm hairs stood on end. (c) Use the idea of conductors and insulators to explain why the iron-roofed stall was struck but the wooden stall next to it was untouched. Support your answer with evidence from your friction and conductivity tests.

(a) All matter is made of atoms with positive protons and negative electrons. Objects become charged when electrons are transferred, for example by friction. The object that gains electrons becomes negative and the object that loses electrons becomes positive. In class, a balloon rubbed with wool and a comb rubbed with cloth attracted torn paper. Two types of charge exist: positive and negative. In the storm cloud, charge separated in a similar way, giving the cloud a large net charge.

(b) The law of charges says that like charges repel and unlike charges attract. We saw two balloons rubbed with wool repel each other, while a wool-rubbed balloon and a nylon-rubbed balloon attracted. The charged cloud affected the charges in the traders' bodies, so the hairs all gained the same type of charge and repelled one another. They spread apart and stood on end.

(c) A conductor has free electrons that move easily. An insulator holds its electrons tightly. In our tests, iron wire and copper wire from a jua kali workshop lit the tester bulb, but sisal twine, dried maize husk and a rubber slipper sole did not. The iron roof is a conductor, so charge could move through it and concentrate there. This gave a low-resistance path for the discharge, which scorched and melted the iron sheet. Wood is an insulator, so charge could not flow through it and the wooden stall was undamaged.

Part 2: Charging Without Touching - Induction and Earthing

The iron roof never touched the storm cloud, yet it gained

(a) When the charged PVC pipe was brought near without touching, the foil leaves diverged. When I touched the ball with my finger while the pipe was nearby and removed the finger before the pipe, the leaves stayed diverged even after the pipe was taken away.

charge. (a) Describe what you observed when a charged PVC pipe was brought near, without touching, the foil electroscope, and what happened when you touched the electroscope ball with your finger and removed the finger before removing the pipe. (b) Using these observations, explain step by step how the iron roof became charged by induction, including the sign of the charge on its upper surface. (c) Explain what earthing is, and how the lightning conductor on the district offices and the drag chain on a fuel tanker on Mombasa Road use it. (d) Explain why the wooden stall could not be charged by induction.	So the electroscope had been left with a lasting charge. (b) The negatively charged cloud was above the roof. Free electrons in the iron were repelled downward, away from the cloud. This left the upper surface closest to the cloud with a positive charge. This is charge separation by induction, with no contact. The charge induced is opposite in sign to the inducing charge, so the upper surface of the roof became positive. The attraction between the cloud and the roof grew, setting up the conditions for the strike. (c) Earthing is giving excess charge a conducting path to the Earth so it drains away. The tanker's drag chain connects the tanker body to the road, so charge from friction does not build up and spark. The lightning conductor on the district offices keeps the building connected to the Earth, so any induced charge drains away safely and the discharge is directed into the ground. (d) Wood is an insulator with no free electrons to move, so charge cannot be redistributed in it. The wooden stall could not gain a charge separation by induction.
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Part 3: Forces Across Empty Space - Coulomb's Law

Charged objects push and pull on each other without touching. (a) Describe what you observed about the force between two charged balloons as their distance changed, and what your F against $1/r^2$ graph showed. (b) Write Coulomb's law and explain what happens to the force if the distance is doubled, and if one of the charges is doubled. (c) Use Coulomb's law to explain why the iron roof, and not the wooden stall, was the structure the cloud acted on most strongly, and why the lightning conductor could intercept the discharge. (d) How does this idea help explain the force acting on charges in the	(a) Two charged balloons repelled each other, and the repulsion became weaker as they were moved further apart. The graph of F against $1/r^2$ was a straight line through the origin, so the force is proportional to $1/r^2$. (b) Coulomb's law: $F = kQ_1Q_2/r^2$, where k is about $9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$. Doubling the distance reduces the force to one quarter. Doubling one charge doubles the force. (c) The force depends on the size of the charges and on the distance. The iron roof, as a conductor, had a significant induced charge, while the wooden stall, an insulator, had almost no induced charge, so its force was much smaller. The roof was also a large conductor close to the cloud, giving a large force. The lightning conductor is tall, which makes r small, and its pointed tip concentrates charge. So it experienced a large attraction and could intercept the discharge safely, even though it was 30 m away. (d) The charges in the hairs were acted on by the charge in the cloud without any contact. The electrostatic force is a force at a distance that acts across the air. It gets stronger as r gets smaller, so the effects were noticed just before the strike.
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traders' hairs and why the force acts across empty space?

Part 4: Storing and Releasing Energy - Capacitors

(a) Describe how you built and tested your foil capacitor, and what you observed when you changed plate area and plate separation. How did the LED show that energy had been stored? (b) Write the definition of capacitance and the formula for the energy stored in a capacitor. Explain, using the Q-V graph, why the energy has a factor of one half and why doubling the voltage has the effect it does. (c) Explain how the storm cloud and the iron roof acted like a capacitor, and use this to explain why the discharge could melt iron sheet. (d) Why does the lightning conductor provide a safer way to release this energy?

- (a) I made a parallel-plate capacitor from two aluminium foil sheets separated by a plastic bag, charged it from a 9 V battery and discharged it through an LED. A larger plate area gave a brighter or longer flash, and a larger separation gave a dimmer or shorter flash. The LED lit after the battery was disconnected, showing that energy had been stored and could be retrieved.
- (b) Capacitance is the charge stored per unit potential difference: $C = Q/V$, measured in farads. The stored energy is $E = 1/2 CV^2$. The Q-V graph is a straight line through the origin. The voltage starts at zero and grows as charge is added, so the work done is the area of a triangle under the graph, not a rectangle. This gives half of QV, and with $Q = CV$ it becomes $1/2 CV^2$. Because V is squared, doubling the voltage makes the stored energy four times larger.
- (c) The cloud base and the iron roof were like the two plates, with air as the insulating dielectric. As the potential difference grew, more charge and energy were stored in the electric field. When the field became strong enough to ionise the air, the capacitor discharged in a rapid bolt. Because the energy depends on V^2 , the huge voltage meant a very large energy was released, enough to scorch and melt the iron sheet and make the radio crackle.
- (d) The lightning conductor gives the stored charge a controlled path to the Earth, so the energy is released into the ground and not through the roof or the people.

Part 5: Putting It Together - Explaining Ol Kalou and Using Electrostatics in Kenya

Write a complete explanation of the Ol Kalou Market lightning strike, in order from the build-up of charge to the final discharge. Your explanation must link charge separation, induction, Coulomb force, conductors and insulators, energy storage and the lightning conductor. Then choose one Kenyan application of electrostatics from the group presentations (for example electrostatic precipitators at

Explanation: Charge builds up in the storm cloud by transfer of electrons, so the cloud holds a large net charge. This charged cloud induces charge on the iron roof: free electrons in the iron are pushed away, leaving a positive charge on the upper surface, with no contact. The cloud and roof then act like the plates of a capacitor with air as the dielectric, and the potential difference and stored energy ($E = 1/2 CV^2$) grow. By Coulomb's law the force depends on the charges and on $1/r^2$, so as the distance falls the force on charges grows, and the field becomes strong enough to ionise the air. This explains the hair standing on end and the radio crackle beforehand. The air breaks down and the stored energy is released in a lightning strike. The iron roof is a conductor with free electrons, so it carried the discharge and melted. The wooden stall is an insulator with no free electrons, so it was untouched. The lightning conductor on the district offices is tall and pointed, concentrates charge, and is earthed, so it gives the discharge a safe path into the ground and protects the building.

Application (example): The electrostatic precipitator at Athi River charges dust particles from cement-making smoke and then attracts them to oppositely charged plates. It uses the same ideas: charge transfer, the law that unlike charges attract, and the electric force acting at a distance. This removes dust from the air and protects the health of the surrounding community.

Athi River, photocopiers in Kisumu, spray-painting booths on Mombasa Road or lightning conductors in Nyandarua) and explain which of the same ideas it uses to protect communities or power technology.	
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the lessons	Cites specific observations from several lessons (balloons, comb, conductivity tests, electroscope, foil capacitor and LED) and connects each to the OI Kalou event.	Cites some evidence from the lessons and links most of it to the OI Kalou event.	Gives little or general evidence, or does not link it to the OI Kalou event.
Scientific accuracy of concepts	Correctly explains electron transfer, the law of charges, conductors and insulators, induction, earthing, Coulomb's law and capacitor energy with correct signs, formulas and units.	Explains most concepts correctly with small errors in wording, signs or units.	Shows major misconceptions, such as charges being created, or confuses the formulas.
Explaining the phenomenon as a cause-and-effect chain	Gives a complete, correctly ordered explanation from charge build-up to discharge that answers all the puzzles: hairs, radio, iron roof versus wooden stall, and lightning conductor.	Gives an ordered explanation that answers most of the puzzles but misses a link.	Lists ideas without a clear order or leaves most puzzles unanswered.
Quantitative reasoning and use of equations	States $F = kQ_1Q_2/r^2$, $C = Q/V$ and $E = \frac{1}{2} CV^2$ correctly and explains effects of doubling quantities, including the Q-V graph area.	States the equations correctly and explains most proportional effects.	Equations are missing or incorrect, or proportional effects are wrong.
Application to Kenyan technology and communication	Clearly links one Kenyan application to the same physics ideas, with clear writing and accurate scientific vocabulary.	Names an application and links it to some of the ideas, with generally clear writing.	Names an application without linking it to the physics, or writing is unclear.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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